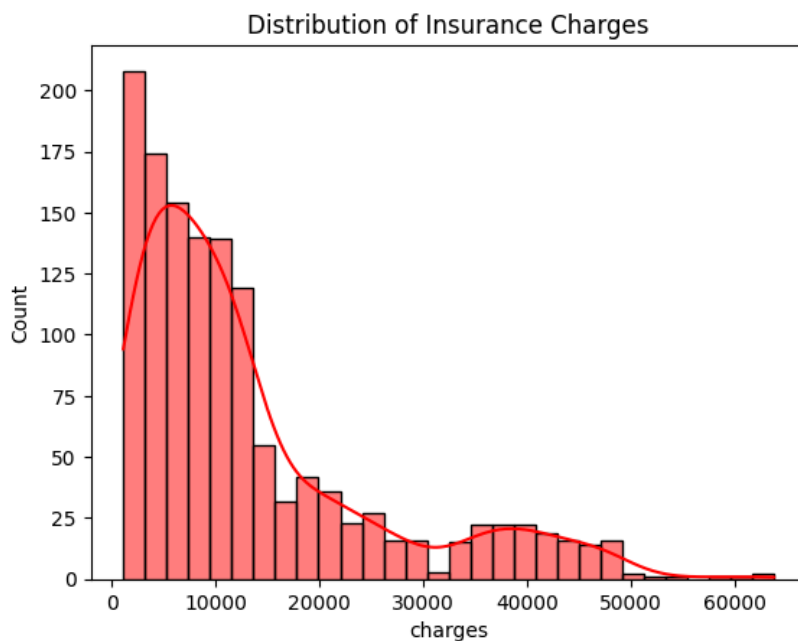


Pricing Data Analysis for HealthGuard Insurance

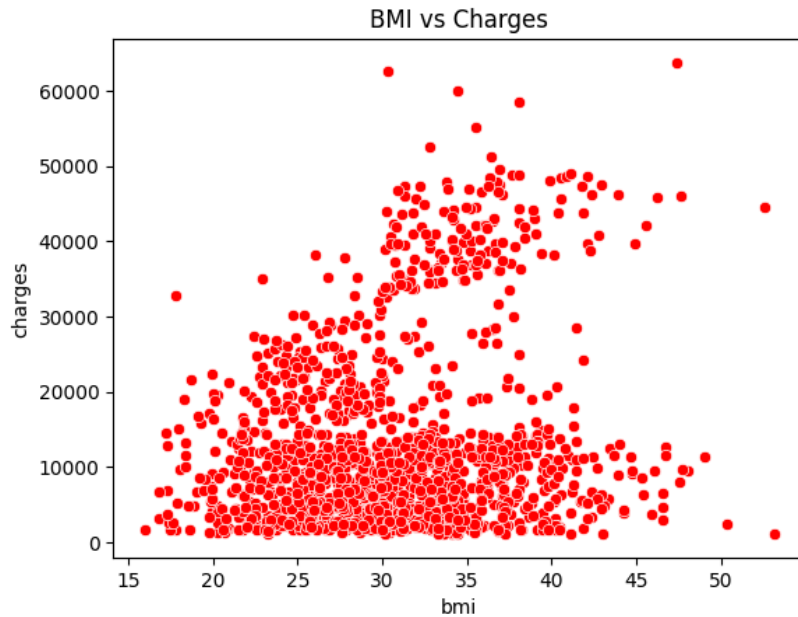
Darren Easler

The dataset presented lists 1,338 customer charges for health insurance as well as their age, sex, number of children, region, BMI, and smoking status. Some of these factors play a much larger role in the charges they pay for health insurance, while others follow a much weaker correlation. Before determining a model to predict the charges a customer may be billed, it is important to understand the correlations of the data. The following analysis looks at the charges based on certain aspects of the customer attributes.

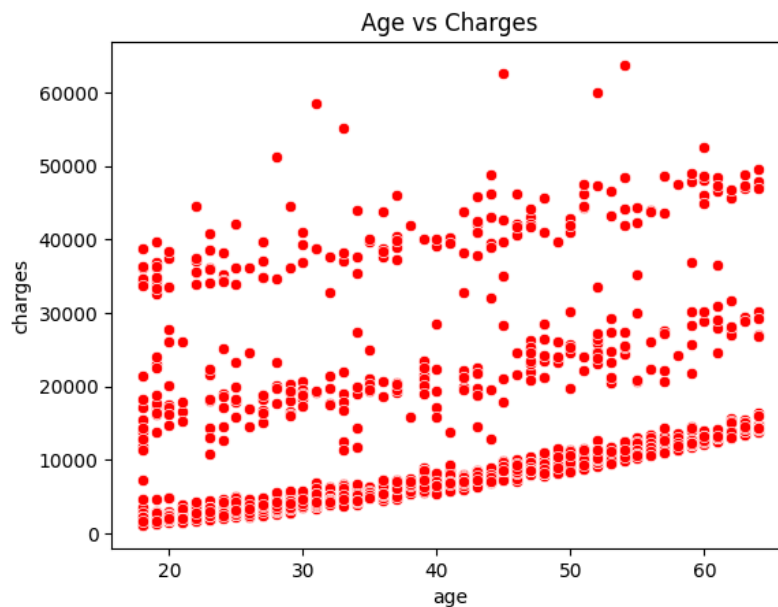


The total charges clearly bunch significantly below \$20,000. The distribution of charges shown is clearly extremely right-skewed, where the vast majority of the pricing exists on the lower extreme of the range.

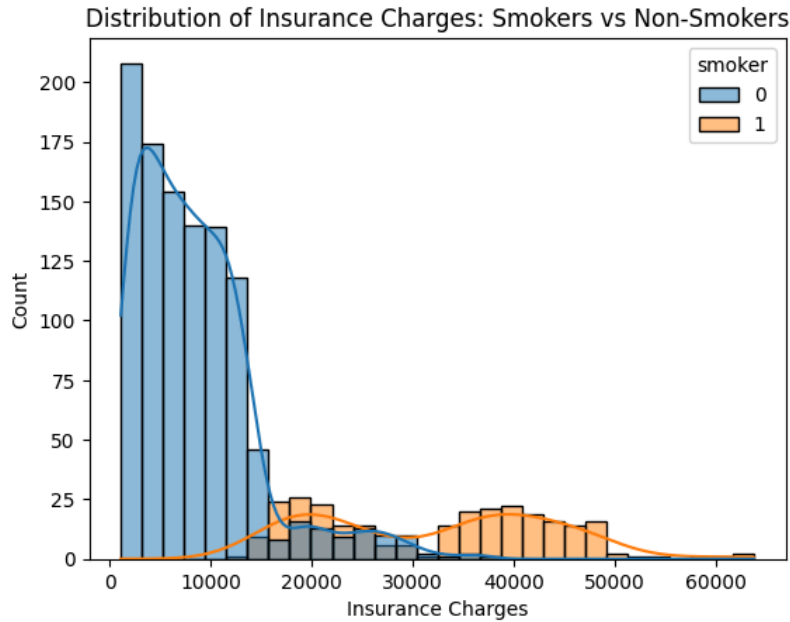
A customer's BMI is a weak predictor of the health insurance charges they will face. Below is the full BMI data compared to the charges, which shows there is weak correlation. For the majority of data, BMI is relatively constant, and even the higher priced data, the scattered covers a majority of the BMI range.



The region and children categories provide very similar correlations, weak and unimpactful to the charges. So, it is best to look at the age and smoking correlations to determine more helpful correlations.

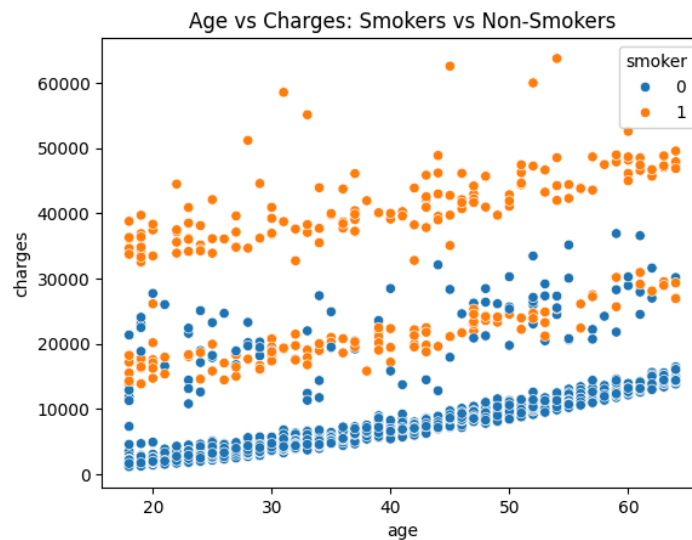


The distinct clusters here show a common trend with age, pricing will *tend to increase with age*. There are three distinct clusters here, each a higher priced band of customers. This likely means that there is another factor that influences the charges beyond simply age. This could be a predictor of these natural bands forming in the data.



The smoking status of the customer is the best predictor of the charge trends, as seen by the graph above. Those who are non-smokers make up almost all of the data under \$25,000, but those who are smokers fall into a category almost entirely over \$20,000 and bunching higher at \$40,000. The trend shows that charges *tend to increase for customers who smoke*.

To see this effect with the natural clusters formed by the age data, combining the past two graphs shows that age and smoking are the most predictable attributes. As age increases, so does cost. The more expensive bands in the age data also correlate to having a higher percentage of smokers as well. From this plot, it is clear that increased age and “yes” for smoking are the highest driver of insurance charges.



These attributes, age and smoking status, will be the most influential in the prediction models for this data.

The first model is a Single Linear Regression. It uses the target 'smoker', which was previously determined to be the most influential feature in predicting the charges. However, neglecting the other features makes the single variable linear regression a poor prediction model for this data, as the other features play a role in the determination of a solid prediction.

The second model is a Multivariable Linear Regression, taking the full feature set into consideration when predicting a charge value. This model significantly improves the learning performance simply by factoring in a scale for all of the features.

The third model is a Polynomial Regression, with a degree of 2. Increasing the degree of polynomial beyond 2 tends to overfit the data, suggesting a parabolic trend of the relationship. To avoid this overfitting, this lower degree polynomial is used as a more complex model without capturing too much of the noise in the data.

The fourth model is a Ridge Regression, which was tuned to have the most appropriate value of alpha; $\alpha=0.1$. This model did not perform as well as the Polynomial Regression, the lack of overfit data neglects the need for stricter regularization.

The fifth model is a Lasso Regression, another regularization model that needs some fine tuning. The alpha value $\alpha=10$ produced the best evaluation metrics, and completely eliminated 'sex' as a predictor for insurance charges. It nearly eliminates 'region' as well, but the predictors 'smoker' and 'age' dominate the model, affirming the predictions made earlier. However, like the Ridge model, the lack of overfit data means that regularization attempts like the Lasso do not significantly improve the model.

The sixth model is a Support Vector Regression (SVR), where three kernels were tested and compared to determine the usefulness of this model. These kernels were linear, poly(deg2), and RBF. The RBF tuned model, with values $C = 1000$ and $\text{Gamma} = 0.1$, produced better evaluation metrics than other regressions, but still not as comparable to the degree 2 polynomial regression. The SVR is very picky to hyperparameter tuning, which limits its overall effectiveness in this context.

The seventh and final model is a Decision Tree Regression. After some testing, the tree of depth 4 proved to be the most effective predictor of the charges, but still not as effective as the polynomial regression tested earlier. The most important aspect of the tree is its ability to be

interpreted with a visual, making it more clear as to why each decision is made, rather than just mathematical predictions from the other models.

Below is a table depiction of the evaluation metrics of each of the tested models.

MODEL	MAE	MSE	RMSE	R_2
Simple Linear	5621.81	52745964.72	7262.64	0.660
Multivariable Linear	4186.51	33635210.43	5799.59	0.783
Poly Regression (2)	2730.32	<i>20530608.94</i>	<i>4531.07</i>	<i>0.868</i>
Ridge	4186.66	33635862.37	5799.64	0.783
Lasso	4187.52	33661029.23	5801.81	0.783
SVR (kernel = RBF)	<i>2434.77</i>	30428021.73	5516.16	0.804
Decision Tree	2697.76	21093484.01	4592.76	0.864

Italicized Values Indicate the Best Metric

More complex models clearly perform better than simpler linear regressions, but the best predictive model is a Polynomial Regression of Degree 2. This model achieved the best balanced performance, and has the best correlation trend to the data. Though it is beaten by SVR for the MAE, which indicates the SVR has a strong average prediction accuracy, the Polynomial Regression wins in RMSE, indicating it makes fewer large prediction errors, and in R-Squared, giving the most consistent predictions with the test data. Higher degree polynomials showed signs of overfitting, so this model is the best polynomial regression of its class.

Final Decision: Polynomial Regression of Degree 2

To flag expensive customers, the Polynomial Regression of Degree 2 model was run with a Logistic Classification. Customers who exceed the median of ‘charges’ are flagged as expensive. The model was split in the same way as the original model, and the following confusion matrix and metric scores were achieved by the model.

Accuracy: 0.933 Precision: 0.960 Recall: 0.903

This model accurately predicts if a customer will be expensive or not 93.3% of the time. This would be an incredibly useful tool in determining the binary split of “expensive” and “not expensive” customers.

The model precisely predicts if a customer will be expensive 96.0% of the time. This model predicts expensive customers to a degree of precision that makes it incredibly significant if it flags a customer as expensive.

The model has a recall percentage of 90.3%, which means that the model “misses” an expensive customer that should be flagged less than 10% of the time. For a larger dataset, these misses become less and less meaningful, as 90.3% will be caught correctly.

	Not Expensive	Expensive
Not Flagged	129	5
Flagged	13	121